

Multimodal integration of metagenomic and clinical data for pathogen identification in lower respiratory tract infection

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Abstract

Distinguishing causative pathogens from incidental microbial detections remains a challenge in metagenomic diagnosis of respiratory infection. We developed AETIA (Agentic Evidence Tracing of Infectious Aetiology), which combines language models and prespecified rules to integrate metagenomic, clinical, microbiological and published evidence. In a retrospective development cohort of 55 patients with suspected lower respiratory tract infection at three hospitals, AETIA achieved 71.6% precision and 91.4% recall against physician-adjudicated pathogens. Its F1 score, summarizing precision and recall, was 80.3%, compared with 14.8% for unfiltered sequencing reports, 50.0% for multiplex polymerase chain reaction and galactomannan testing, and 45.5% for culture. In a 41-patient subset, AETIA exceeded four language models directly prompted with the same data by 13.6–15.1 F1 points. Each organism’s selection or exclusion was linked to the underlying evidence. These findings support integrating clinical context with metagenomic results for pathogen attribution; external validation and prospective evaluation are needed to establish clinical utility.

Introduction

Lower respiratory tract infection is frequently treated without a microbiological diagnosis. In a large prospective study of adults admitted with community-acquired pneumonia, no pathogen was identified in 62% despite extensive conventional testing¹. In critically ill patients, previous

antibiotic treatment further reduces culture sensitivity², noninfectious inflammatory syndromes can resemble pneumonia³, and the differential diagnosis extends to fungi, viruses and opportunistic pathogens. Metagenomic next-generation sequencing (mNGS) was developed to address this gap by detecting bacterial, fungal, viral and parasitic nucleic acids without a prespecified target^{4,5}. Clinical studies have shown its diagnostic potential in meningoencephalitis⁶, bloodstream infection^{7,8} and lower respiratory tract infection^{3,9–11}, and the assay is increasingly used in critically ill patients in Asia^{2,12}. Its breadth, however, carries a clinical risk: tests that detect incidentally carried microbes can lead to antimicrobial overuse and adverse outcomes¹³.

The main challenge is interpretation rather than detection. Respiratory mNGS identifies upper-airway flora^{14,15}, reagent and environmental contaminants¹⁶, herpesviruses shed or reactivated during critical illness^{17,18} and airway-colonizing *Candida*¹⁹, as well as causative pathogens. In a prospective bronchoalveolar-lavage cohort, mNGS identified a definite or probable pathogen in 65% of patients with lower respiratory tract infection, compared with 20% by culture, but had a specificity of 75.4% and a positive predictive value of 78.5% against a composite reference standard¹⁰. Distinguishing infection from colonization therefore remains a central limitation^{5,11}. Read thresholds and subtraction of no-template controls reduce background signal but do not resolve the biological ambiguity of a colonized respiratory specimen^{20,21}.

Resolving that ambiguity requires evidence from outside the sequencing assay. Culture establishes viability and enables susceptibility testing^{22,23}, multiplex PCR provides rapid, validated target-specific results²⁴, galactomannan contributes an independent criterion for invasive fungal disease²⁵, serology can establish infection after the organism has cleared from the airway[cite], and imaging defines the anatomical syndrome of pneumonia²². Each result must be read against the specimen and its timing: infection confined to blood, tissue, an abscess or an empyema may be absent from respiratory fluid, and intermittent or treated infection may be missed outside its detection window^{21,26,27}. Determining causation consequently requires integration of sequencing, conventional microbiology and clinical context.

Computational approaches to this problem have so far drawn only on the sequencing data. A rules-based model that scored organisms by relative abundance and known pathogenicity separated pathogens from commensals in tracheal aspirates from critically ill adults³, and classifiers adding host gene expression improved diagnosis of lower respiratory tract infection in critically ill children¹³ and of sepsis from plasma²⁸. Their host-response components require RNA sequencing of host transcripts, and none incorporates culture, antigen or serological results or the clinical course that physicians use to adjudicate causation.

Large language models could perform this integration because they encode clinical knowledge²⁹, generate differential diagnoses from vignettes^{30,31} and have been evaluated as infectious-diseases consultants³². Their performance on curated tasks, however, has not consistently transferred to clinical records. In an evaluation of 2,400 patient cases, current models performed worse than physicians, did not reliably follow diagnostic or treatment guidelines, had difficulty interpreting laboratory results and were sensitive to both the quantity and the order of the information they received³³. Model outputs also vary between runs and may not reveal which evidence determined a prediction³⁴. A generated explanation cannot by itself establish fidelity because a model’s stated reasoning may not correspond to the computation that produced its

answer³⁵; in clinical applications, such post-hoc explanations can provide false reassurance³⁶. Multi-agent systems^{37–39} and retrieval augmentation^{40,41} address parts of this problem, but clinical use also requires reproducible decisions, traceable evidence and an explanation written separately from the prediction. Evaluation must also guard against leakage⁴²: clinical notes often record the treating team’s final diagnosis, which a model can copy rather than infer. To our knowledge, no approach has combined mNGS with conventional microbiology and the clinical record to attribute causation organism by organism under these constraints.

Here we present AETIA (Agentic Evidence Tracing of Infectious Aetiology), which integrates respiratory mNGS with conventional microbiology and clinical data. Modality-specific language-model agents extract structured evidence; deterministic rules select candidate pathogens; and case-matched PubMed evidence can rescue candidates not selected initially. The clinical rationale is written only after the selected list has been fixed. In patients with suspected lower respiratory tract infection from three medical centres in Taiwan, we compared AETIA with individual diagnostic read-outs and with direct prompting of general-purpose language models against physician-adjudicated pathogens. AETIA identified the adjudicated pathogens more accurately than any single read-out or directly prompted model, reported far fewer organisms than the sequencing report while retaining most true pathogens, and linked each selection and exclusion to the evidence behind it. The system is intended to provide an evidence-linked candidate list for physician review, not an autonomous diagnosis.

Results

AETIA separates evidence extraction, rule-based selection and literature-grounded rescue

AETIA processes nine clinical sources for each patient: admission diagnosis, underlying conditions, blood counts, other laboratory values, cultures, FilmArray, galactomannan, molecular microbiology and imaging, together with the laboratory mNGS report (Fig. 1a). Seven modality-specific language-model agents, one each for imaging, clinical history, laboratory values, galactomannan, PCR, culture and the mNGS report, convert these sources to a fixed schema. Each agent receives one modality and no reference labels. A deterministic summary combines the outputs into organism-specific evidence and host context.

Selection is rule-based (Fig. 1b). Each organism on the mNGS report is graded by its signal relative to negative controls, relative and absolute read burden, within-specimen dominance, anatomical alignment and support from hospital tests. These features and host immunosuppression determine a five-level causative ranking (Methods and Supplementary Tables 1 and 2). Prespecified guardrails demote common colonizers, oral flora, environmental organisms, skin flora, low-actionability herpesviruses, weak mould signals and non-specific laboratory labels. Organisms in the top two levels are selected; intermediate organisms require protected-pathogen status or direct support. A language-model safety review flags unselected organisms for further assessment but cannot alter or reorder the selected list.

For each selected or flagged candidate, the pipeline assembles literature, sequencing and clinical evidence (Fig. 1c). Up to ten PubMed articles are assessed first for species-specific evidence

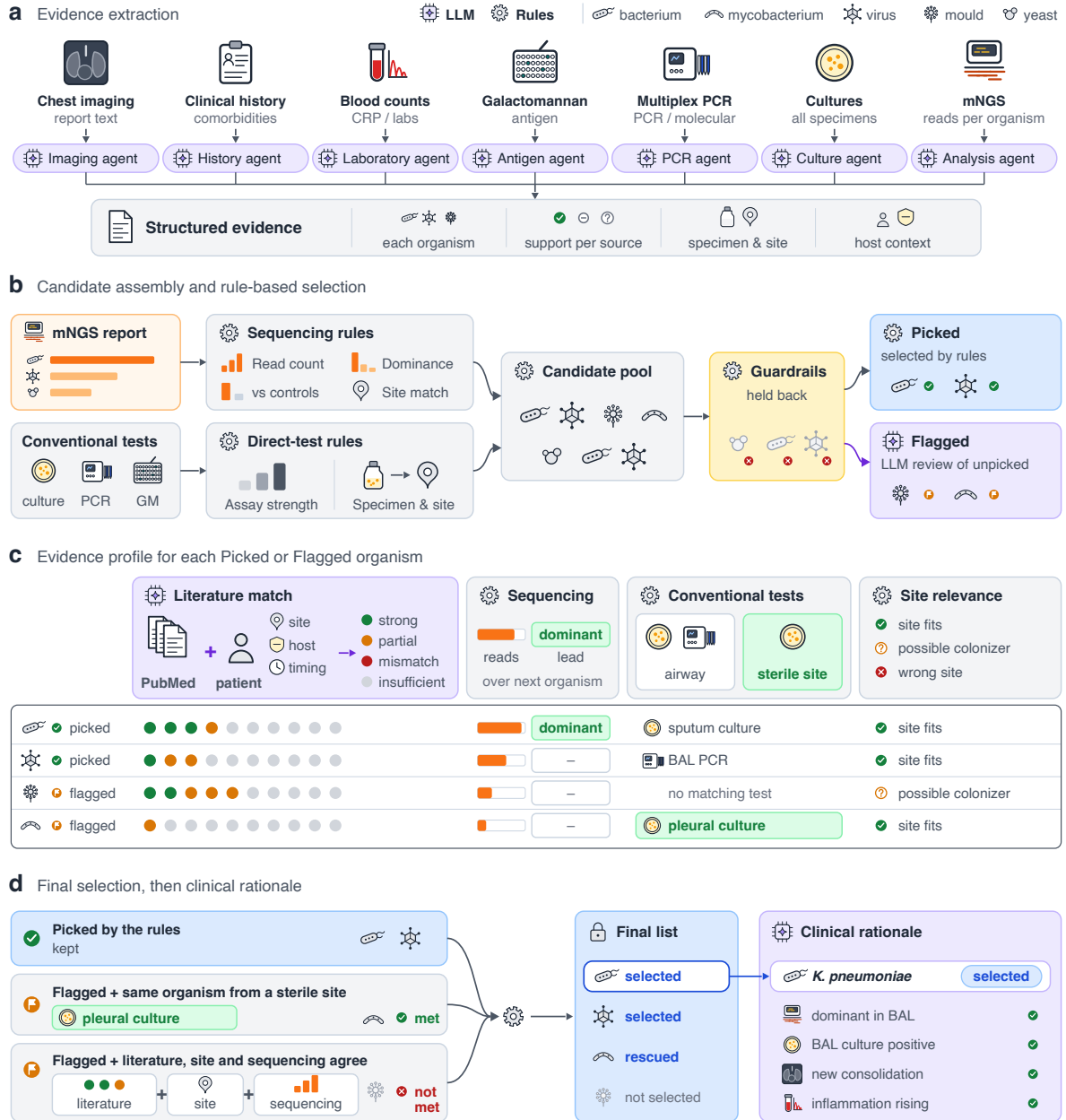


Figure 1: **Overview of AETIA.** Violet chip, language-model step (extracts or judges evidence, or writes the clinical rationale; never selects); cog, deterministic rule. One schematic patient with seven detections is followed through all panels. **a**, Seven modality-specific language-model agents turn the record into structured evidence for each organism. **b**, Rules grade mNGS detections and conventional-test results, merge them into one candidate pool and hold back likely colonizers; a language-model safety review flags unpicked organisms but cannot select them. Two organisms are Picked, two Flagged and three held back. **c**, Each Picked or Flagged organism receives an evidence profile, not a decision: one verdict per PubMed article matched to the patient's site, host and timing; read count and dominance over the next organism; conventional tests by specimen class; and site relevance. Bars are schematic. **d**, Picked organisms are kept. A Flagged organism is added only if the same organism is isolated from a sterile site or tissue (here, a mycobacterium in pleural-fluid culture) or if literature, site and sequencing all agree (not met by the mould). The list is fixed before a language model writes the clinical rationale; the *K. pneumoniae* entry is drawn from a prototype vignette (Fig. 3e) and is not a delivered record. Criteria are defined in Methods and Supplementary Tables 1–4.

of infection at the target site and then for similarity to the patient’s specimen, syndrome, host, phenotype and timing. Separate modules quantify absolute and relative sequencing strength, exact-organism support from non-sequencing tests and colonization risk. Missing or pending tests remain unknown. The final rule retains every deterministic selection and adds a candidate only when the same organism is isolated from a sterile site or tissue, or when case-matched literature, site alignment and a qualifying sequencing signal are all present (Fig. 1d). The clinical rationale is written only after this list has been fixed. The delivered table and traceable files report the evidence and rule associated with each selected and excluded organism (Fig. 3e).

AETIA preserves pathogen recall while improving precision over individual tests

The primary evaluation included 55 patients with suspected lower respiratory tract infection from Kaohsiung Medical University Hospital (KMUH, $n = 30$), Taipei Veterans General Hospital (TVGH) and Tri-Service General Hospital (TSGH; combined $n = 25$), drawn from the 105-specimen registry (Table 1; Extended Data Fig. 1). At sampling, 71% were in intensive care, 56% had hospital- or ventilator-associated pneumonia, 37% had an active malignancy, 39% were receiving systemic corticosteroids and 60% had bronchoalveolar-lavage specimens. Treating infectious-diseases physicians provided the reference pathogen lists. Performance was evaluated at the exact-species (patient, organism) level using micro-averaged precision, recall and F1.

AETIA achieved 71.6% precision, 91.4% recall and 80.3% F1 (Fig. 2a,b). The unfiltered mNGS report had similar recall (93.1%) but 8.0% precision (F1 14.8%). FilmArray and galactomannan testing had 67.6% precision and 39.7% recall (F1 50.0%), whereas culture had 48.1% precision and 43.1% recall (F1 45.5%). Thus, AETIA retained most of the sensitivity of the unfiltered report while increasing precision approximately ninefold.

AETIA achieves higher recall than directly prompted general-purpose models

We compared AETIA with four general-purpose models (GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra) in 41 patients (KMUH $n = 30$, TVGH and TSGH $n = 11$). Each model received the same de-identified mNGS and clinical data as AETIA, one expert-written prompt and a structured output schema, without retrieval, a supplied candidate list or reference labels (Methods and Supplementary Note 2). Predictions were stored before evaluation.

The four directly prompted models had precision of 67.7–70.5%, recall of 56.6–60.5% and F1 of 62.4–63.9% (Fig. 2c,d). In the same 41 patients, AETIA achieved 67.7% precision, 90.8% recall and 77.5% F1 (69 true positives, 33 false positives and 7 false negatives), exceeding the directly prompted models by 13.6–15.1 F1 points. The difference was attributable mainly to recall: GPT-6 Astra had the highest recall among the directly prompted models, but remained 30.3 percentage points below AETIA. [Slide 3 gives 71.62% precision here; the frozen tally TP 69 / FP 33 gives 67.65%. Resolve.]

AETIA records supporting evidence for 123 selections and reasons for 395 exclusions

The source records were multimodal and longitudinal. Among 32 KMH admissions with available source workbooks, each admission contained a median of 247 recorded results (IQR 214–263; range 157–423) across laboratory testing, imaging and microbiology during a 31-day period, with new results on a median of 16 days and 1,452 words of free text (Fig. 3a–c). The records named a median of eight organisms per admission (IQR 6–11). Of 270 organism–admission pairs, 81% were named by only one read-out and 91% were never isolated from a sterile site. Among 365 chest-imaging studies, 90% mentioned pneumonia, infiltrate or consolidation, but 44% of these reports qualified the finding (Fig. 3d). Culture reports with no growth were issued after a median of 6 days, compared with 3 days for reports with growth.

Across the 55-patient evaluation, the deterministic rules considered 518 candidates and selected 114; the final evidence rule added nine, yielding 123 selected candidates (2.2 per patient) and 395 exclusions (Fig. 3h). These nine additions could arise either from an exact-organism sterile-site or tissue isolate or from case-matched literature combined with site alignment and a qualifying sequencing signal. Each delivered record reports the selection or exclusion rule, supporting and opposing findings, specimen site and timing, sequencing burden and dominance, and any matched PubMed articles (Fig. 3e–g). A candidate absent from the mNGS report can enter through an exact-organism hospital test at a relevant site; an atypical pathogen supported only by IgM serology is retained for review. [Count the hospital-evidence selections among the 123 from the delivered records.]

Two prototype cases illustrate how the recorded evidence distinguishes respiratory colonization from invasive infection (Fig. 3e,f; Supplementary Note 3). In one ventilated patient, mNGS identified dominant *Klebsiella pneumoniae* (68,569 reads) and *Candida albicans*. New bilateral consolidation, pleural effusions, increasing C-reactive protein, hypoxaemia and a dominant sequencing signal supported *K. pneumoniae*; low, non-dominant reads, low-burden respiratory culture, absence of a sterile-site isolate and the respiratory-*Candida* guardrail supported exclusion of *C. albicans*. In a second patient, multiple blood cultures grew *Candida tropicalis* and bronchoalveolar reads were high, satisfying the direct-evidence criterion despite the guardrail. [Vignettes are from the January 2026 prototype; regenerate from the delivered pipeline outputs.]

For all cases, organism selection, the rule path and the structured evidence values were fixed before the clinical rationale was generated. The generator received the completed decision, and the audit required one entry for each selected organism, prohibited additional organisms and blocked incomplete output. This procedure establishes structural consistency with the completed decision; the clinical usefulness and record-level fidelity of the clinical rationales were not independently evaluated.

Airway mNGS reports contain more organisms but have limited corroboration across tests

The three-hospital registry contained 105 mNGS specimens (Table 1; Extended Data Fig. 2). Airway specimens contained a median of five reported organisms (IQR 3–7; $n = 67$), compared with one (IQR 1–2; $n = 38$) in blood, cerebrospinal fluid, tissue and other sterile-site speci-

mens. Apparent differences by illness severity disappeared after restriction to bronchoalveolar lavage: the median was five organisms in intensive care and on the ward, 5.5 with corticosteroid treatment and five without it, and five with and without acute respiratory distress syndrome. The difference between hospital- or ventilator-associated and community-acquired pneumonia persisted within lavage specimens (five versus three organisms). Thus, the number of organisms requiring interpretation was associated primarily with sampling an airway compartment.

Conventional tests corroborated a minority of reported organisms (Extended Data Fig. 2b). Of 414 organisms on the 105 mNGS reports, 75 (18.1%) had a same-species culture during the admission; conversely, 90 of 165 cultured organisms (54.5%) were absent from mNGS. Among 312 registry comparisons, 49.0% were sequencing detections without a culture counterpart, 29.8% were partial matches, 6.7% were mismatches and 5.8% were exact matches, using the registry's categories. Viruses accounted for 27.1% of reported organisms and 21.8% of adjudicated pathogens, and 100 of 112 reported viruses were herpesviruses.

Adjudicated pathogen lists were available in the shared workbook for a separate subset of 33 registry cases (Supplementary Note 6). Their mNGS reports contained 104 organisms, of which 38 (37%) were adjudicated as causative and 66 (63%) as colonizing, contaminating or incidental. Read count ranked a causative organism above a non-causative organism in only 61% of within-case pairs, and the distributions overlapped across five orders of magnitude. Ten species were causative in one patient and non-causative in another. For example, *Acinetobacter nosocomialis* with 189,467 reads in lavage fluid was adjudicated as colonization, whereas *Pneumocystis jirovecii* with 7 reads was adjudicated as causative.

The same 33-case analysis contained 55 adjudicated pathogens, 17 (31%) of which were absent from the mNGS report of the sequenced specimen. Another read-out identified 16 of these 17 pathogens: culture in eight, PCR in five, galactomannan in two and IgM serology in one. Fourteen were identified within two days of mNGS sampling, and 11 were identified in a different specimen type. Sequencing and conventional tests therefore had complementary failure modes.

The registry also contained 22 specimens (21.0%) without a recorded causative pathogen. Six had an explicit negative entry, whereas 16 had the dash used by one hospital for unlabelled rows. mNGS reported at least one organism in 18 of these 22 specimens, with 47 organisms in total and as many as eight in one specimen. [Confirm with the TVGH investigators whether the dash in the adjudication column means “no causative pathogen identified” or “not recorded”; if the former, these 22 specimens are the no-infection stratum the evaluation cohorts lack, and the baselines and AETIA should be scored on them.]

Nearly half of patients had treatment changes attributed to mNGS without a detectable survival difference

Antimicrobial treatment changed in 69 of the 105 registry patients (65.7%), and clinicians attributed a change to mNGS in 52 (49.5% of the registry; 58.4% of the 89 patients with recorded attribution). These changes comprised 36 escalations and 13 de-escalations; 13 of all 19 de-escalations were attributed to mNGS. The cohort was severely ill: 82.7% underwent mechanical ventilation within three days of sampling and 48.6% died in hospital. Reports associated and not

associated with an attributed treatment change contained the same median number of organisms (four).

No survival difference was detected between the 52 patients whose treatment change was attributed to mNGS and the 37 whose change was not. In-hospital death occurred in 30 of 52 patients (57.7%) and 21 of 37 patients (56.8%), respectively (odds ratio 1.04, 95% confidence interval 0.45–2.41; $P = 1.00$, Fisher’s exact test). Adjustment for intensive care at sampling, acute respiratory distress syndrome, active malignancy, corticosteroid treatment and age gave a similar estimate (adjusted odds ratio 1.22, 95% confidence interval 0.46–3.21; $P = 0.69$; 51 deaths among 89 patients). The two groups were closely matched before the change (acute respiratory distress syndrome 51.9% versus 54.1%, active malignancy 40.4% versus 40.5%, corticosteroid treatment 40.4% versus 48.6%), and the single imbalance ran against the group whose change was attributed to mNGS (intensive care 71.2% versus 91.9%). Patients whose treatment changed for any reason died more often than patients whose treatment did not, but they were also more severely ill beforehand (acute respiratory distress syndrome 56.5% versus 25.0%, corticosteroid treatment 44.9% versus 25.0%); that comparison measures the indication for changing treatment rather than its effect and we do not interpret it. This retrospective attribution was not an assigned exposure, every patient underwent mNGS and the registry contained neither antimicrobial agents nor administration dates; the analysis therefore does not estimate a causal effect of mNGS-guided treatment or the appropriateness of therapy.

Early defervescence and improved oxygenation also occur in patients who subsequently die

We examined physiological trajectories in 48 KMH admissions with machine-readable complete discharge records (31 deaths and 17 survivors), using every dated laboratory value and vital sign and anchoring observations to the day of the mNGS specimen (Extended Data Fig. 3; Supplementary Methods). The analysis was descriptive because the records did not contain antimicrobial start dates, stop dates or coverage of the adjudicated pathogen.

Among the 31 patients who died, death occurred a median of 15 days after the mNGS specimen (IQR 7–21; range 1–57) (Extended Data Fig. 3a); 22 died more than 7 days and 17 more than 14 days after sampling. Every patient who was febrile in the two weeks before sampling—10 decedents and 5 survivors—was afebrile on days 3–7. Among decedents, the proportion with a maximum temperature of at least 38 °C fell from 39% before sampling to 18% on days 0–2 and 3% on days 3–7. Temperature fell by a median of 0.5 °C between the pre-sampling and day 3–7 windows ($n = 27$ paired; $P = 0.001$, Wilcoxon signed-rank). Arterial P_{O_2} increased by days 3–7 in 17 of 27 decedents. These changes were often transient: fever returned in 9 of 21 decedents whose temperature improved, and P_{O_2} subsequently fell in 12 of the 17 whose oxygenation improved.

Platelet trajectories differed descriptively between survivors and decedents (Extended Data Fig. 3c). Median platelet count in survivors increased from 169 before sampling to 257 on days 3–7 and 360 on days 8–14, whereas the corresponding values in decedents were 82, 48 and $55 \times 10^3 \mu\text{L}^{-1}$. Among 14 assessable decedents, the count declined by a median of $19 \times 10^3 \mu\text{L}^{-1}$ from the pre-sampling to day 3–7 window ($P = 0.004$) and improved in 2. Lactate, white-cell

count, creatinine and procalcitonin did not show corresponding differences between the groups. These trajectories show that early defervescence and improved oxygenation were not confined to survivors; without treatment timing or coverage, they cannot establish whether the causative organism was correctly identified or treated.

Discussion

Respiratory mNGS findings required organism-level clinical interpretation. In the registry, 63% of reported organisms were not adjudicated pathogens, read count ranked a pathogen above a non-pathogen in only 61% of pairs and ten species were causative in some patients but not in others. Clinicians nevertheless attributed an antimicrobial change to mNGS in 49.5% of the registry, including 13 of 19 de-escalations. The patient’s subsequent course did not indicate whether the organism treated was the right one: every KMH patient febrile before the specimen was afebrile by day 3–7 and oxygenation improved in 17 of 27 patients who nevertheless died, so the bedside signals read as confirmation of an empirical choice occurred in survivors and decedents alike and reversed in roughly half. In the 55-patient evaluation cohort, AETIA combined modality-specific extraction, deterministic scoring, safety review and case-matched literature to achieve 91.4% recall and 71.6% precision. It retained the sensitivity of the unfiltered mNGS report while substantially improving precision and outperformed individual conventional read-outs. On 41 patients, it also exceeded four general-purpose language models prompted directly with the same data by 13.6–15.1 F1 points. These findings support structured integration of sequencing, conventional microbiology and clinical data rather than interpretation by a single assay threshold or model prompt.

The comparison with direct prompting suggests that the main advantage was task structure rather than model capacity. The direct prompts specified colonization, anatomical-site and negative-control considerations, and their precision was similar to that of AETIA. Their lower recall indicates that general caution was applied too broadly. A deterministic rule can encode organism-specific exceptions, such as lower read thresholds for protected pathogens or additional requirements for environmental organisms. The literature stage then evaluates whether published cases match the species, site, syndrome and host. [Replace the proposed *Stenotrophomonas* and *Pneumocystis* examples with verified cases from the per-patient comparison.] The small recall difference among the four directly prompted models further suggests that increasing model capacity alone may not resolve this formulation problem.

Traceability is a separate contribution. The clinical records contained many discordant and asynchronous results: a median of 247 recorded results and eight named organisms per admission, with only 19% of organism–admission pairs supported by more than one read-out. AETIA records the evidence supporting or opposing every selected organism and the reason each non-selected organism was excluded. Because the structured decision is fixed before the clinical rationale is generated, the rationale can be checked against the rule path and source values. This differs from a model-generated account produced in the same pass as the prediction, which may not reflect the computation that determined the answer³⁵ and can provide false reassurance in clinical use³⁶. This design does not establish that the rationales are adequate for clinical use, but it makes discrepancies identifiable and testable.

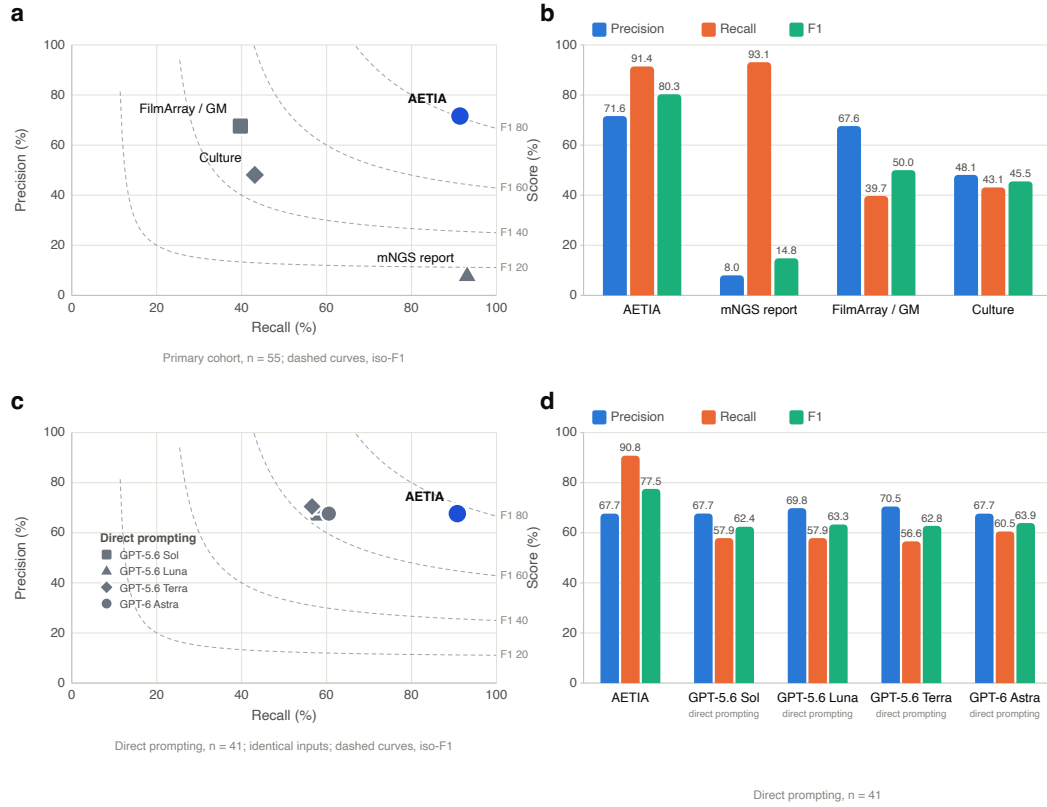


Figure 2: Pathogen selection compared with individual read-outs and directly prompted language models. **a**, Precision against recall for AETIA (blue) and for each read-out taken at face value: all organisms in the mNGS report (triangle), all positive FilmArray or galactomannan targets (square) and all cultured organisms (diamond); dashed curves, constant F1. **b**, The same comparison as precision, recall and F1. $n = 55$ patients (KMUH 30; TVGH and TSGH 25); unit, (patient, organism) pair at species level; reference standard, physician-adjudicated pathogens (Methods). **c**, Precision against recall for AETIA and four general-purpose language models prompted directly with the same de-identified mNGS and clinical record; marker shapes identify the models. **d**, The same model comparison as precision, recall and F1. $n = 41$ patients (KMUH 30; TVGH and TSGH 11); micro-averaged over (patient, organism) pairs. Each model received one prompt and a JSON schema, without retrieval or reference labels (Methods; Supplementary Note 2). AETIA values are from the frozen decisions on the same patients (TP 69, FP 33, FN 7).

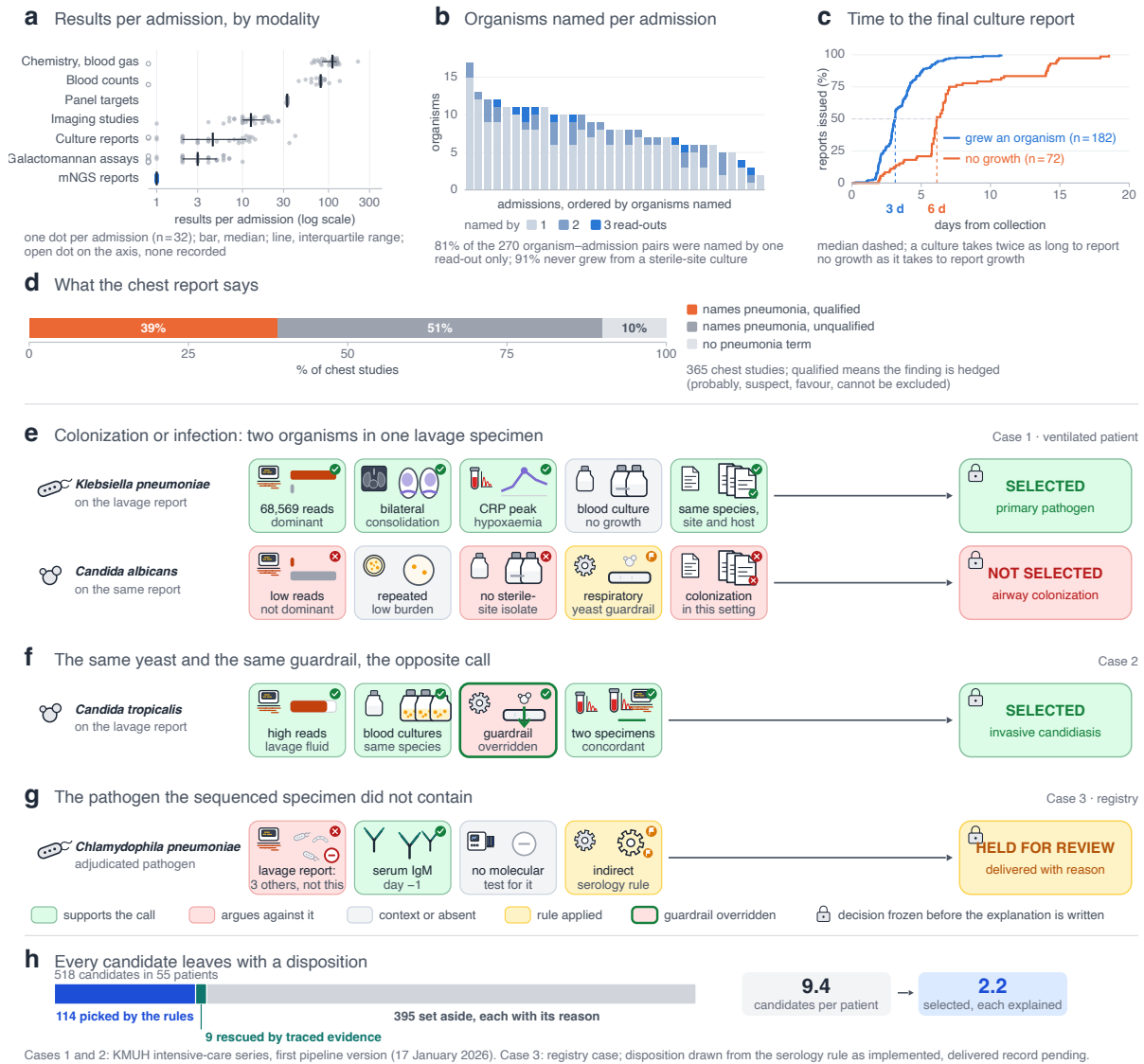


Figure 3: Clinical evidence underlying individual pathogen-selection decisions. **a–d**, Analysis of 32 KMH admissions with available source workbooks; admissions sequenced more than once were counted once. **a**, Number of recorded results per admission by modality, compared with the single mNGS report (median 247, IQR 214–263; new results recorded on 16 of 31 days). **b**, Organisms named per admission (median 8, IQR 6–11). Each bar represents one admission and shading indicates the number of read-outs that named an organism. Of 270 organism-admission pairs, 81% were named by one read-out and 91% were never isolated from a sterile site. **c**, Cumulative percentage of 254 culture reports issued by day and outcome. The median reporting time was 6 days for no growth and 3 days for growth. **d**, Wording of 365 chest-imaging reports: 39% qualified a finding of pneumonia, infiltrate or consolidation, 51% reported such a finding without qualification and 10% used none of these terms. Qualifying terms are defined in the Supplementary Methods. **e–g**, Delivered records for three cases, with one glyph per evidence source and the decision and supporting evidence fixed before the clinical rationale was written. **e**, *Candida albicans* classified as airway colonization and *Klebsiella pneumoniae* selected from the same lavage specimen. **f**, *Candida tropicalis* selected because blood cultures grew the same species (overridden guardrail, ringed). **g**, Registry pathogen absent from the sequenced specimen and retained for review on the basis of IgM serology. **h**, Disposition of 518 organisms in 55 patients: 114 selected by the deterministic rules, 9 added by traced evidence and 395 excluded with a recorded reason. [Panels e and f use the two prototype vignettes of the 2026-01-17 talk; regenerate from the delivered pipeline records and obtain case-level approval.]

Table 1: **Characteristics of the three-hospital mNGS registry.** The 105-specimen registry from which the 55- and 41-patient evaluation cohorts were drawn (Extended Data Fig. 1). Values are n (%) or median (interquartile range).

Characteristic	All patients ($n = 105$)
<i>Hospital</i>	
KMUH	51 (48.6%)
TVGH	35 (33.3%)
TSGH	19 (18.1%)
Age, years	69 (56–75)
Male sex	68 (64.8%)
Current or former smoker	19 (18.1%)
Intensive care unit at sampling	75 (71.4%)
<i>Underlying conditions</i>	
Diabetes mellitus	29 (27.6%)
Chronic kidney disease	16 (15.2%)
End-stage renal disease on dialysis	4 (3.8%)
Liver cirrhosis	1 (1.0%)
Autoimmune disease	6 (5.7%)
Active malignancy	39 (37.1%)
Systemic corticosteroids [†]	40 (38.5%)
ARDS on the day of sampling [†]	48 (46.6%)
<i>Pneumonia category</i>	
Community-acquired	21 (20.0%)
Hospital- or ventilator-associated	59 (56.2%)
Not classified	25 (23.8%)
PaO ₂ /FiO ₂ ratio at sampling	200 (152–302)
<i>mNGS specimen</i>	
BALF	63 (60.0%)
Blood	24 (22.9%)
Sputum	4 (3.8%)
CSF	6 (5.7%)
Tissue	5 (4.8%)
Other	3 (2.9%)
Organisms per mNGS report, median (IQR)	4 (1–6)
mNGS report with no organism	8 (7.6%)
Any positive culture	69 (65.7%)
<i>Clinical course</i>	
Antimicrobial therapy changed	69 (65.7%)
Escalation	49 (46.7%)
De-escalation	19 (18.1%)
Change attributed to mNGS [‡]	52 (49.5%)
Mechanical ventilation within 3 days of sampling [†]	81 (82.7%)
In-hospital death [†]	51 (48.6%)

[†]Denominator excludes patients without a recorded value. [‡]Denominator is all patients; attribution was not recorded for 16 patients. KMUH, Kaohsiung Medical University Hospital; TVGH, Taipei Veterans General Hospital; TSGH, Tri-Service General Hospital; BALF, bronchoalveolar lavage fluid; CSF, cerebrospinal fluid.

The study has several limitations. It was retrospective, included 55 patients in the primary analysis and 41 in the model comparison, and used data from three hospitals in one country and one sequencing laboratory. The reference standard was physician adjudication informed by the mNGS report and was therefore not independent of the index test. The rule set was developed with access to these labels, no patient was held out for external validation and the reported performance is a development-set estimate. Both evaluation cohorts required at least one adjudicated pathogen at a pulmonary site, so patients whose final diagnosis was infection elsewhere or no infection were excluded. This is a selection on the outcome: it defines a population that cannot be identified when the test is ordered, and it removes the patients in whom the site and colonization guardrails would be most consequential, since the correct output for them is an empty pulmonary list. The precision reported here therefore applies to patients in whom pneumonia was ultimately confirmed and should not be read as the false-positive rate a clinician would encounter when sending mNGS on suspicion. This missing stratum is clinically relevant: 22 of 105 registry specimens had no recorded causative pathogen, although the interpretation of 16 unlabelled entries requires confirmation. Scoring AETIA and the comparators in these patients is an important next analysis.

The pipeline also shares constraints with the underlying evidence. An organism detected only in a non-sterile specimen cannot be proven causative without corroborating clinical or microbiological findings. Guardrails that correctly remove most airway colonizers may also demote uncommon true infections, including *Candida* pneumonia, herpes simplex virus tracheobronchitis or galactomannan-negative aspergillosis. Conversely, 31% of adjudicated pathogens were absent from the mNGS report of the sequenced specimen; the pipeline can recover such pathogens only when another test identifies them. These errors require an additional specimen or assay, not greater sequencing depth. The literature and case-fit stages use a commercial language model and a changing PubMed corpus. Frozen article snapshots permit replay of the reported analysis, but future runs may retrieve different evidence and model judgements are not deterministic. The two illustrative cases originated from an earlier 35-patient series and require confirmation using the delivered records from the primary cohort. Finally, we assessed agreement with adjudication rather than clinical benefit. We did not determine whether the selected lists changed antimicrobial treatment or outcomes, nor did independent physicians assess the completeness, fidelity or usefulness of the clinical rationales. The registry does record what the clinicians did next, and acting on the sequencing report was not associated with survival in it; that null is not evidence against benefit, because every registry patient was sequenced, the exposure is the treating team’s own retrospective attribution rather than an assignment, and no antimicrobial agent or date is recorded from which the appropriateness of a change or the time to appropriate therapy could be computed. The higher mortality among patients whose treatment changed measures the indication for changing it rather than its effect, and any endpoint measured in days is subject to immortal time, because a change credited to a specimen taken days into an admission can only be recorded in patients who survive to receive it. The trajectory analysis sharpens this gap without closing it. It establishes that the window is long—death followed the specimen by a median of 15 days—and that the response signals available at the bedside do not discriminate, but it is observational, every patient was sequenced, and antipyretics, corticosteroids,

teroids, sedation and ventilator settings act directly on temperature and gas exchange and were not recorded in a form permitting adjustment. The temperature column of the whole-record extracts also required repair before use (Methods).

A prospective study should freeze the rules and prompts before enrolment, include patients without infection, use an adjudication committee blinded to the pipeline output and evaluate changes in antimicrobial treatment and clinical outcomes. Two analyses of the existing cohort should precede it. The KMH discharge records contain the antimicrobial charts, so the interval from the report to a change of therapy and the adequacy of that change against the adjudicated pathogen are extractable without enrolling anyone; and the 22 specimens without a recorded pathogen provide the no-infection stratum once their adjudication convention is confirmed. The quality of the clinical rationale should be assessed independently by comparing decisions made with and without the evidence record and by auditing exclusion reasons in false-negative cases. The same architecture could incorporate resistance-gene evidence and be evaluated in blood-stream and central nervous system infections. Such extensions should follow external validation of the present pathogen-selection rules.

Methods

Ethics and study oversight

This retrospective study analysed existing clinical records and mNGS reports; patients were not contacted, and pipeline outputs were not used in care. The institutional review boards of Kaohsiung Medical University Hospital (KMH; [protocol number]), Taipei Veterans General Hospital (TVGH; [protocol number]) and Tri-Service General Hospital (TSGH; [protocol number]) approved the study and waived informed consent for retrospectively collected, de-identified data. Direct identifiers were removed before records were sent to a commercial language-model service through an application programming interface whose terms exclude training on submitted data, and reference labels were withheld from all model inputs. In the direct-prompting payloads, calendar dates were replaced by days relative to the mNGS specimen, a check blocked any payload that retained an exact date, and provider-side storage of responses was disabled. [IRB coverage of external API use; de-identification standard; date handling in the AETIA stages; zero data retention.] AETIA was evaluated as research software returning candidate pathogens for physician review.

Study design and participants

This retrospective, multicentre diagnostic-accuracy study is reported according to STARD 2015⁴³ and TRIPOD-LLM⁴⁴ [attach both checklists]. It included patients who underwent mNGS for suspected lower respiratory tract infection at three tertiary centres in Taiwan (KMH, TVGH and TSGH) between August 2023 and September 2025. The index test was AETIA and the reference standard the adjudicated causative-pathogen list. Comparators were conventional read-outs taken at face value and four general-purpose language models prompted directly.

The hospitals' shared registry held 105 mNGS specimens (KMH, 51; TVGH, 35; TSGH, 19) with demographics, comorbidities, microbiology, mNGS reports, treatment changes and out-

comes (Table 1). Membership of the two evaluation cohorts was recorded in versioned manifests (Extended Data Fig. 1). The primary cohort comprised 55 patients (KMUH, 30; TVGH and TSGH, 25) with a complete standardized record, a per-specimen mNGS report and at least one adjudicated pathogen at a pulmonary site. Patients were excluded if chest imaging, blood counts or microbiology were missing or too poorly documented for multimodal evaluation, or if the final adjudicated diagnosis was not pneumonia (upper respiratory tract infection alone, infection at another site or a noninfectious condition). The estimates therefore apply to patients in whom lower respiratory tract infection was ultimately confirmed, a population that cannot be identified when the test is ordered (Discussion). [Numbers excluded per reason and alternative diagnoses; whether exclusions preceded model inference; reasons for the two-hospital steps 54 → 48 → 25.] The direct-prompting cohort comprised the 30 KMUH patients and the 11 TVGH and TSGH patients whose records were complete in every section ($n = 41$).

Metagenomic sequencing and conventional microbiology

mNGS was ordered by the treating physicians and performed by one clinical laboratory (Asia Pathogenomics, New Taipei City, Taiwan) with a published workflow^{45,46}: nucleic-acid extraction without host depletion; 75-bp single-end sequencing on an Element Biosciences AVITI system (about 40 million reads per sample); quality filtering^{47,48}; removal of reads aligning to T2T-CHM13^{49,50}; and alignment to a curated microbial database, with organisms reported when their normalized abundance exceeded ten times the highest value in a concurrent negative control (Supplementary Methods). Specimens were mainly bronchoalveolar lavage fluid (60%) and blood (23%). AETIA used only the clinical report (reported organisms, read counts and control-subtracted ranks), not raw reads.

Culture, the BioFire FilmArray Pneumonia Panel²⁴, the Platelia *Aspergillus* galactomannan assay⁵¹ and targeted molecular tests were performed in each hospital’s accredited laboratory and taken as reported (Supplementary Methods). Within the pipeline, a FilmArray bacterial target contributes evidence in proportion to its semi-quantitative bin, and galactomannan is positive at an index of ≥ 1.0 in lavage fluid and ≥ 0.5 in serum²⁵.

Reference standard

A committee of three infectious-diseases physicians from the participating centres adjudicated each patient, as in clinical mNGS studies^{6,20}. It reviewed the presentation, imaging, blood counts, inflammatory markers, conventional microbiology, mNGS report, clinical course and response to treatment. It then classified every organism detected by any method as a causative pathogen of the current episode or as a colonizer, contaminant, reactivated virus or bystander, using clinical context for organisms that culture cannot confirm. Patients had one to eight causative pathogens. [Decision rule and resolution of disagreement; independent labelling and κ ; timing relative to pipeline development.] Two label revisions affecting four patients were made against the source adjudication documents after every method’s predictions had been stored, and all methods were rescored without re-running any model. [Who revised the labels; whether blind to predictions.] Labels were available only to the evaluation scripts. Because the committee saw the mNGS report, the reference standard was not independent of the index test

(Discussion).

Data extraction and standardization

Rule-based parsers split the KMH per-patient workbooks and the TVGH and TSGH master workbook into clinical sections and extracted text from case-presentation slides. Content present only in images was transcribed from rendered slides, which were linked to registry cases by patient, date and file metadata. A language model (GPT-5, OpenAI; temperature 1) converted each section to a fixed schema for nine domains: admission diagnosis, underlying conditions, complete blood count, other laboratory values, cultures, FilmArray, galactomannan, molecular microbiology and imaging. The mNGS report was the tenth input. Observations were restricted to a window anchored on the mNGS specimen, records without a parsable time were excluded, and hospital-specific identifiers prevented collisions between institutions.

Overview of AETIA

AETIA analyses one patient at a time in five stages (Fig. 1): language-model extraction of evidence by modality; a rule-based initial selection; a language-model safety review of unselected organisms; evidence assessment of the retained candidates and a rule-based final selection; and a clinical rationale written after the list is fixed. Language models extract or assess evidence; version-controlled rules make every selection. Each model task used a fixed prompt and output schema (Supplementary Table 5). For every organism considered, the pipeline recorded the evidence for and against selection, the rule that determined its disposition and SHA-256 hashes of the inputs.

Modality-specific evidence extraction

Seven language-model agents (GPT-5) each received one part of the record, without access to the other agents or the labels: imaging; clinical history and underlying conditions; blood counts and other laboratory values; galactomannan; FilmArray and other molecular microbiology; culture; and the mNGS report, whose agent assigned each organism to its source specimen and anatomical site.[The code runs five clinical agents plus an mNGS-to-specimen agent with a deterministic alternative (`analyze_llm_agent.py:56-111`); reconcile, or state the configuration of the reported runs.] The host agent mapped immunosuppression, malignancy, corticosteroid exposure, renal failure and related factors to prespecified vulnerability categories. The imaging agent graded the likelihood and pattern of infection. Imaging was used as context, never as organism confirmation, which required culture, FilmArray, galactomannan or molecular evidence. A rule-based summary then recorded, for each organism, the support from each modality (support, no support or unknown), the anatomical alignment of its specimen and the host context. Severe host vulnerability widened consideration of borderline organisms but could not by itself cause selection.

Initial pathogen selection

Each reported organism was described by its read count, relative abundance in the run, control-subtracted rank, dominance within the specimen and the anatomical relevance of that specimen (Supplementary Table 1). Repeated detections were resolved by rank and read burden, and names were normalized with a fixed synonym dictionary that did not merge distinct species. The scorer combined these features with specimen alignment, direct hospital-test support and host context to assign one of five ordered causative levels (Supplementary Table 2). Strong support required a favourable control-subtracted rank with high abundance or within-specimen dominance. Intermediate signals could be strengthened by exact-organism evidence from another test or by pathogen-specific rules, and weak or anatomically discordant signals were downgraded.

Guardrails addressed respiratory *Candida*, oral and aspiration flora, skin and environmental organisms, reactivated or low-specificity viruses and weak non-dominant detections (Supplementary Table 3). Strongly supported organisms were selected. Borderline organisms needed compatible clinical context, direct microbiological support or a pathogen-specific rule, never host vulnerability alone, and respiratory *Candida* always required evidence of invasive infection. When nothing qualified, the strongest candidate that passed the exclusion rules was reported as the best available. An organism absent from the mNGS report could be selected on exact-species culture, FilmArray, galactomannan or targeted molecular evidence from a relevant specimen. IgM serology alone was insufficient: *Chlamydophila* or *Mycoplasma pneumoniae* supported only by IgM was retained for contextual review.

Safety review of unselected organisms

A language-model review (GPT-5, reasoning effort medium) examined the unselected organisms, given the deterministic result, the structured hospital evidence and the specimen assignments. It could not remove or reorder a selection. Its prompt stated that immunocompromise or membership of a clinically important group was insufficient on its own and restated the rules for colonization, reactivation, oral and environmental flora and *Candida* (Supplementary Note 1). Rules then standardized names and removed candidates supported only by anatomically irrelevant specimens or weak context. The review could therefore enlarge the set sent to evidence assessment but could not add an organism to the final list.

Evidence assessment and final selection

Each initial selection and each candidate retained by the review was assessed on five dimensions (Supplementary Table 4 and Supplementary Methods). For the literature dimension (module A), PubMed was searched for the organism with lower-respiratory-tract terms (publications to 30 June 2026; up to ten articles), and a language model (GPT-5.6 Luna) judged each abstract for human evidence, with a traceable excerpt, that the exact species causes infection at the relevant site. With too few assessable articles, the module abstained. For case matching (module A1), the model compared the same articles with a label-blind patient summary that omitted selections, read counts, culture and PCR results. Rules graded each comparison as a strong or partial match, a mismatch or insufficient evidence; colonization, a conflicting site or syndrome, non-human

evidence or broader-taxon evidence could not support rescue. Sequencing strength (module B) captured absolute burden and relative prominence in the specimen. Direct microbiological evidence (module C) was graded by assay specificity and anatomical relevance, highest for exact-species detection at a sterile site or in tissue; respiratory *Candida* culture alone did not count. Anatomical coherence (module D) was recorded for audit, and an explicit anatomical mismatch could not be overridden. Modules B–D were computed alongside A1, not as sequential filters.

The final rule (R5) kept every initial selection and added an unselected organism by either of two routes: exact-species evidence from a sterile site or tissue, or a prespecified level of case match with anatomical alignment and either high absolute burden or strong relative prominence (thresholds in Supplementary Table 4). Rescued organisms were appended after the initial selections without ranking. Failing both routes was recorded as insufficient evidence for selection, not as evidence against infection.

Clinical rationale and delivery

After selection, a language model (GPT-5.6 Luna) wrote one rationale per selected organism from the fixed decision, which it was told was immutable, and from candidate-specific microbiology within two days of the mNGS specimen[template default; take the window from the run manifest]. Automated checks required exactly one entry per selected organism, no organism outside the list and every required field; a failure blocked delivery. The delivered record gave the disposition of every organism detected by any method, with each statement linked to its structured evidence (Fig. 3). For a selected organism, it gave the route, supporting and opposing findings, literature, sequencing evidence, specimen site and timing. For a non-selected organism, it gave the exclusion rule or unmet criterion and stated that non-selection does not exclude infection.

Development of rules and prompts

The investigators developed the rules, thresholds, guardrails and prompts iteratively on the patients reported here, with labels available and no patients held out. They were kept under version control, and the reported decisions came from one frozen rule version (Supplementary Note 4). The direct-prompting comparators used one prompt version throughout (Supplementary Note 2). [Developers; iterations and patients consulted; freeze date; whether the direct-prompting prompt was refined on these patients; relative development effort.]

Comparators

Conventional read-outs were scored at face value, without reinterpretation: every organism on the mNGS report, every positive FilmArray or galactomannan target and every organism cultured within the evidence window. [Check against the scoring script; handling of normal flora, no growth and pending results.]

To test whether performance came from the pipeline or from access to a language model alone, four general-purpose models (GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra; OpenAI) were prompted directly in the 41-patient cohort. Each received one request

containing the nine standardized sections and the mNGS report (organisms, read counts, control-subtracted ranks and specimen types), without identifiers or any field derived from the reference standard or from pipeline decisions. The Traditional Chinese prompt cast the model as an infectious-diseases reviewer and stressed that detection does not establish causation. It named colonization, anatomical site, *Candida*, environmental organisms and reactivation as concerns and allowed zero or more species-level pathogens. It required supporting and opposing evidence, excluded organisms and limitations (Supplementary Note 2). Models were queried once per patient (medium reasoning effort, strict JSON output schema, at most 4,000 output tokens, no tools, retrieval or conversation history), and responses were stored with their hashes and model identifiers before scoring. After a correction to the linkage between patients and records, 22 patients were re-queried under unchanged settings (Supplementary Methods); no response was edited.

Outcome measures and scoring

The unit of evaluation was the (patient, organism) pair. Names were normalized with a fixed synonym table (for example, CMV to human cytomegalovirus) and compared as sets at exact-species level. Genus and syndrome labels did not match, so *Candida* against *Candida tropicalis* gave one false positive and one false negative. The primary outcomes were micro-averaged precision, recall and F1. Secondary outcomes were patient-level macro-averages (an empty prediction scoring zero precision), the proportions of patients with at least one, all or exactly the reference pathogens, and the number of patients with no prediction (Supplementary Note 5). The selection made by the initial rules alone was scored in the same way, isolating the organisms added by the final rule. Because both cohorts required an adjudicated pathogen, precision does not capture false positives in pathogen-negative patients. Rationales were checked for completeness and for consistency with the decision but were not rated for usefulness or faithfulness [reader study].

Secondary registry analyses

Four exploratory analyses, not used to develop or evaluate the rules, describe the evidence available before integration: the concordance of sequencing with conventional microbiology (Supplementary Note 6), the composition and timing of the clinical record (Fig. 3a–d), physiological trajectories (Extended Data Fig. 3), and treatment changes and outcomes (Extended Data Fig. 2). Their definitions, counts and tests are given in the Supplementary Methods. Only aggregate or de-identified per-detection data left the protected records.

Computational reproducibility

An orchestrator without selection logic validated each stage’s input, ran the stages in a fixed order, recorded commands and output hashes and never overwrote outputs. Reported decisions were regenerated from archived intermediate files, and direct-prompting metrics from stored responses. Replaying the inputs through the current release reproduced the selected set in all 55 patients; causative levels changed in 15 and selection order in 5 (Supplementary Note 4). [Per

stage: dated model snapshot, query dates, schema failures and retries; tokens, cost and time per patient.] The test suite runs offline without credentials.

Statistics and reproducibility

No statistical method was used to predetermine sample size; the cohorts included every eligible registry patient when the datasets were finalized. All methods were evaluated on the same records, so there was no randomization. Labels were withheld from the language-model stages and read by the evaluation scripts only after predictions were stored. The rule developers, however, had access to the labels and no patients were held out, so the reported performance is a development-set result (Discussion). The clinical records are not public and cannot have been used to train the evaluated models.

Because organisms cluster within patients, 95% confidence intervals for the micro-averaged metrics were estimated by resampling patients (10,000 bootstrap replicates, percentile intervals). Differences in F1 from each comparator were estimated by paired bootstrap, and exact-set agreement was compared by McNemar’s exact test. The comparison with the mNGS report at face value was primary; the other six comparisons were secondary and unadjusted for multiplicity. Interval widths are reported in place of a power calculation, and results are also given separately for KMH and for TVGH and TSGH. Rule-based stages were exactly reproducible. Language-model stages were run once for the reported results, and run-to-run variability was estimated from repeat runs with unchanged inputs: the range of F1 and the proportion of patients with identical selections. [Compute the intervals, tests, stratified metrics, ablation and repeat runs (state n); until then, metrics are point estimates.]

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The patient-level records, mNGS reports, reference labels and model inputs and outputs contain protected health information and cannot be shared publicly. Aggregate cohort characteristics are given in Table 1, and source data for Fig. 2 are provided with this paper. De-identified per-patient evaluation tables (predicted and reference organism sets for every method) and the PubMed identifiers and article judgements from the literature analyses will be provided as Supplementary Data. Requests for patient-level data should be addressed to the corresponding author and will be reviewed by [data-access committee; response time]; access requires approval by the three institutional review boards and a data-use agreement that prohibits re-identification.

Code availability

The AETIA pipeline, including all rules, prompts, configuration files, synthetic example data and the offline test suite, is available at <https://github.com/OliveWoo/Multimodal-Diagnosis-Model> (commit d82c0f7 for this manuscript). Reproducing the

language-model stages requires OpenAI and PubMed access; the deterministic stages and the replay of the reported decisions do not. The code will be released under [licence] and archived at Zenodo ([DOI]). The scripts that generate this manuscript’s figures and tables are at <https://github.com/ythuang0522/aetia-paper>.

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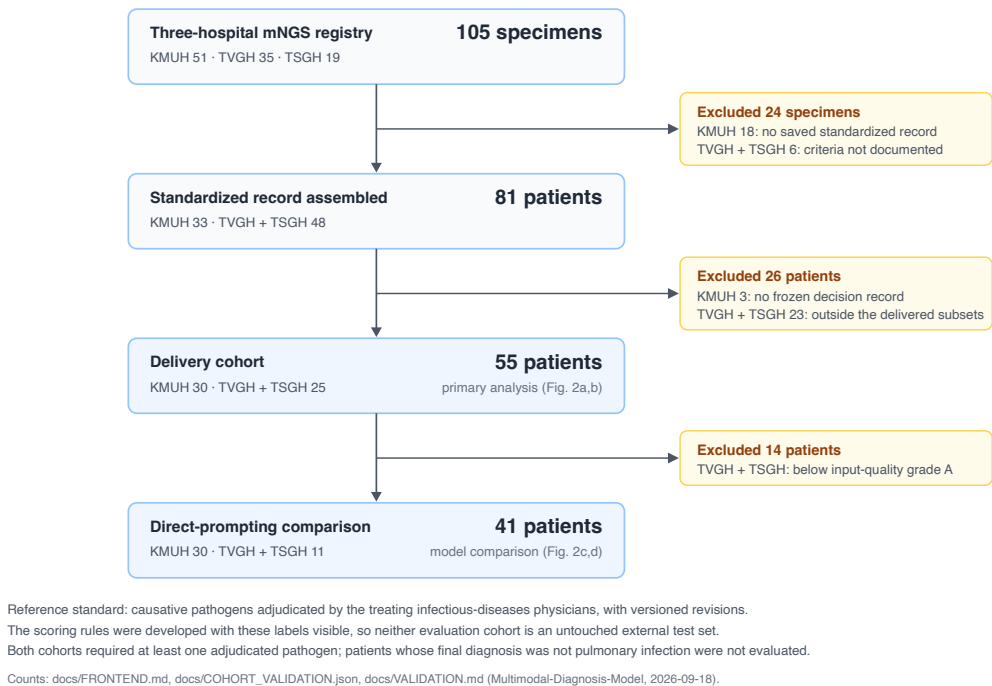
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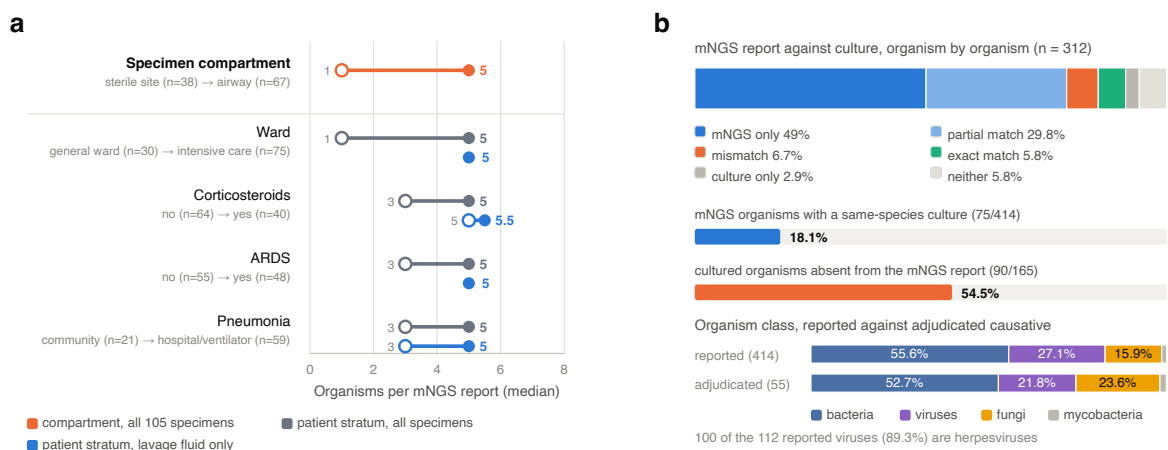
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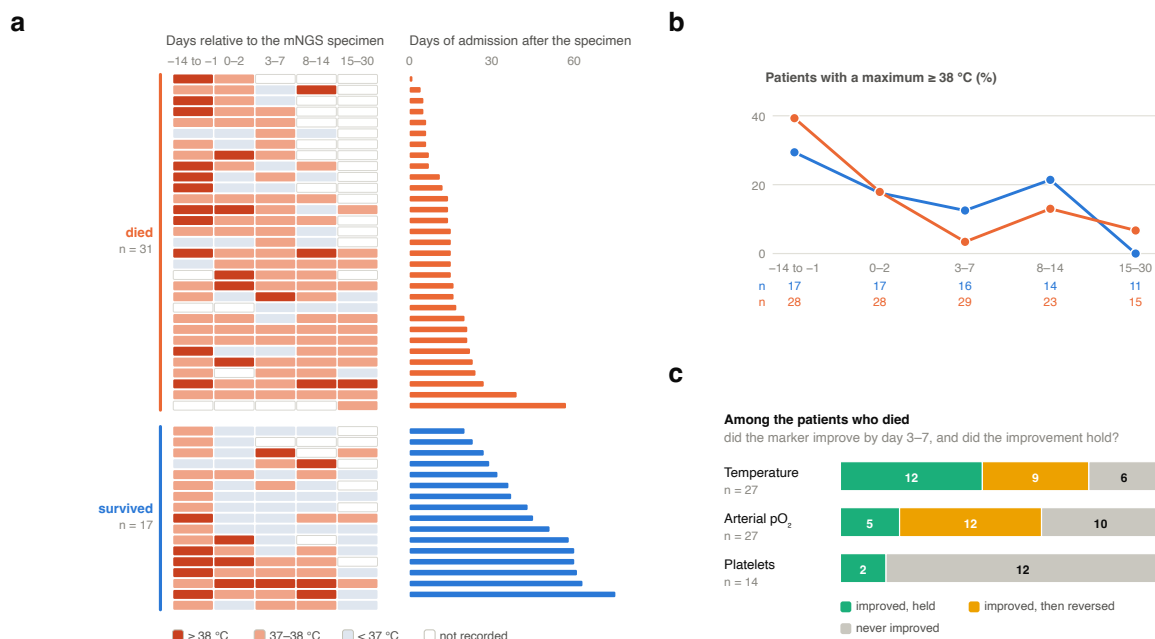
Extended Data



Extended Data Fig. 1: **Cohort flow.** A single pooled flow from the three-hospital mNGS registry to the two evaluation cohorts, with the contribution of each hospital given at every node and exclusions branching to the right. The two hospital lines differ in where their adjudication is stored and in how their standardized records were assembled, which is a matter of data provenance rather than of eligibility (Methods). Amber, exclusion steps. Patients were excluded for insufficient essential clinical data or a final adjudicated diagnosis other than pneumonia (Methods); the allocation of these reasons to the two-hospital steps remains incompletely documented [state the 54 → 48 → 25 reasons, and the number of patients excluded for each reason]. Counts from the repository’s cohort validation of 2026-09-18.



Extended Data Fig. 2: **What the registry’s mNGS reports contain, and how they compare with culture.** The three-hospital registry of 105 mNGS specimens (Table 1). **a**, Organisms per mNGS report (median). Each row is a dumbbell joining the two levels of one contrast; the open circle is the lower level, the filled circle the higher, and the number beside each circle is that level’s median. Orange, the specimen compartment (sterile-site against airway specimens, all 105); grey, a patient stratum over all 105 specimens; blue, the same stratum restricted to bronchoalveolar lavage. The ward, corticosteroid and ARDS gaps present over all specimens close once the compartment is held fixed, so the length of the report tracks the specimen and not the severity of illness; only the pneumonia-category difference persists. **b**, Agreement between the mNGS report and culture. Top, the registry’s own organism-by-organism comparison (312 comparisons; the workbook’s category names are used unchanged, and it does not define partial match or mismatch). Middle, the two corroboration rates, with numerator and denominator given. Bottom, organism class among the 414 reported organisms and the 55 adjudicated causative pathogens. Source: `tables/make_impact.py`.



Extended Data Fig. 3: **The course after the sequencing report, under conventional care.** The 48 KMH admissions with complete machine-readable discharge records (31 deaths, 17 survivors); day 0 is the mNGS specimen. No patient in this analysis received AETIA: these are the trajectories of routine care that motivate the work. **a**, One row per admission, grouped by outcome and ordered by length of admission. Left, the highest temperature recorded in each window after the specimen; right, the days of admission that followed it. The temperature block empties from left to right in both groups: fever resolves whether or not the admission ends in death. **b**, The same temperatures as the proportion of patients reaching 38°C , with the number contributing to each point. **c**, Counted per patient rather than per window, for the patients who died: whether each marker improved by day 3-7 and whether the improvement held to day 8-30. Temperature and oxygenation improved in most and reversed in about half; platelets, which tracked the outcome, improved in 2 of 14. Windows are -14 to -1, 0-2, 3-7, 8-14 and 15-30 days. Antipyretics, corticosteroids, sedation and ventilator settings act on these variables and are not adjusted for; no causal claim is made (Supplementary Methods).